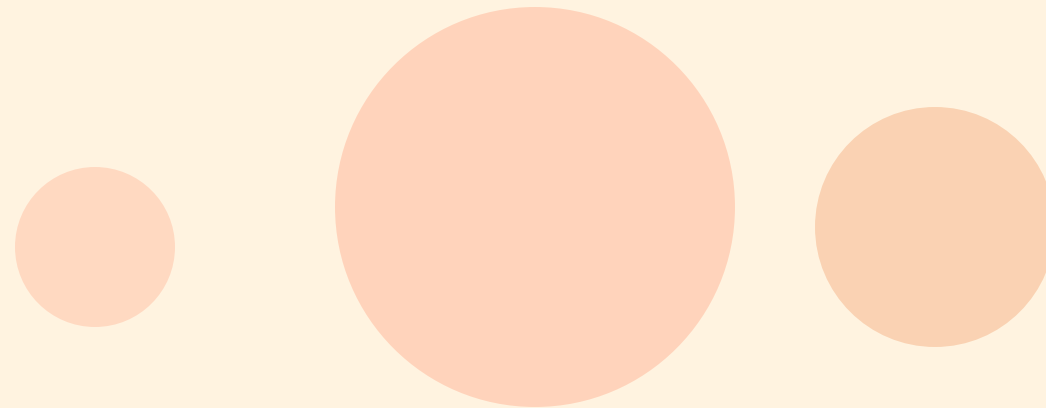
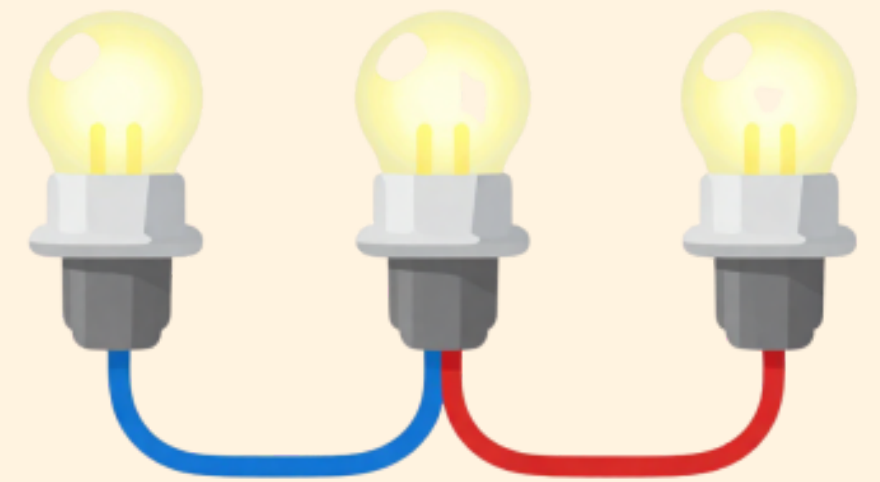


Parallel Circuit

Independent LEDs



What We'll Learn Today

1. What is a parallel circuit?
2. How LEDs work independently
3. Build our own parallel circuit!
4. Compare with series circuits

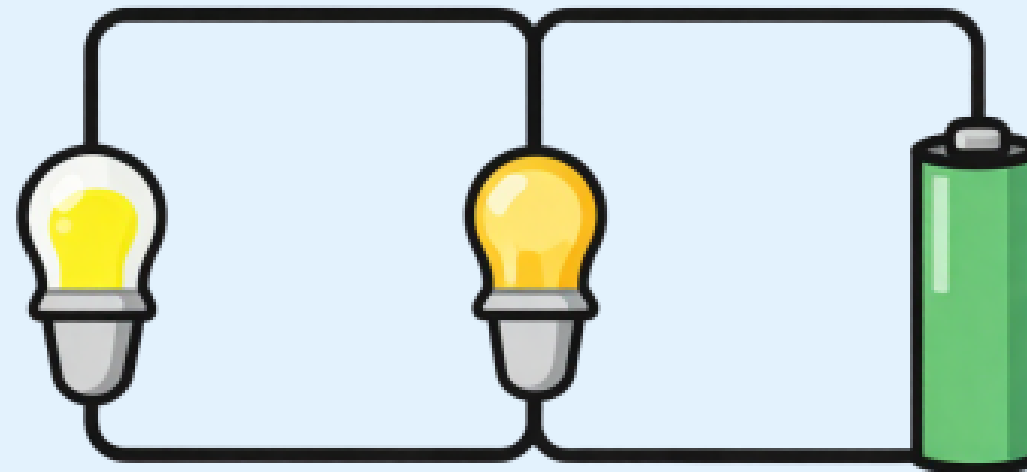


Quick Recap: Series Circuit

In a series circuit, all components are connected in ONE path.

If one LED breaks, ALL LEDs go off!

Like old Christmas lights.

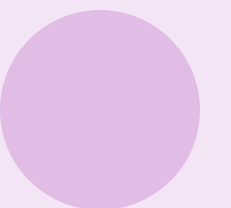
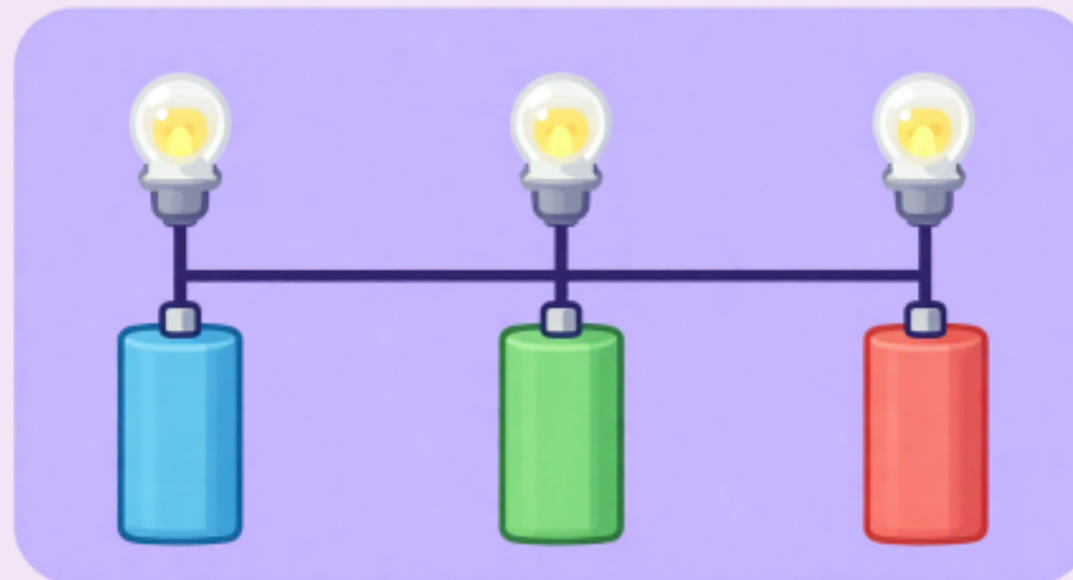


What is a Parallel Circuit?

A parallel circuit has MULTIPLE paths for electricity to flow.

Each LED gets its own separate path from the battery.

Electricity can choose different routes!

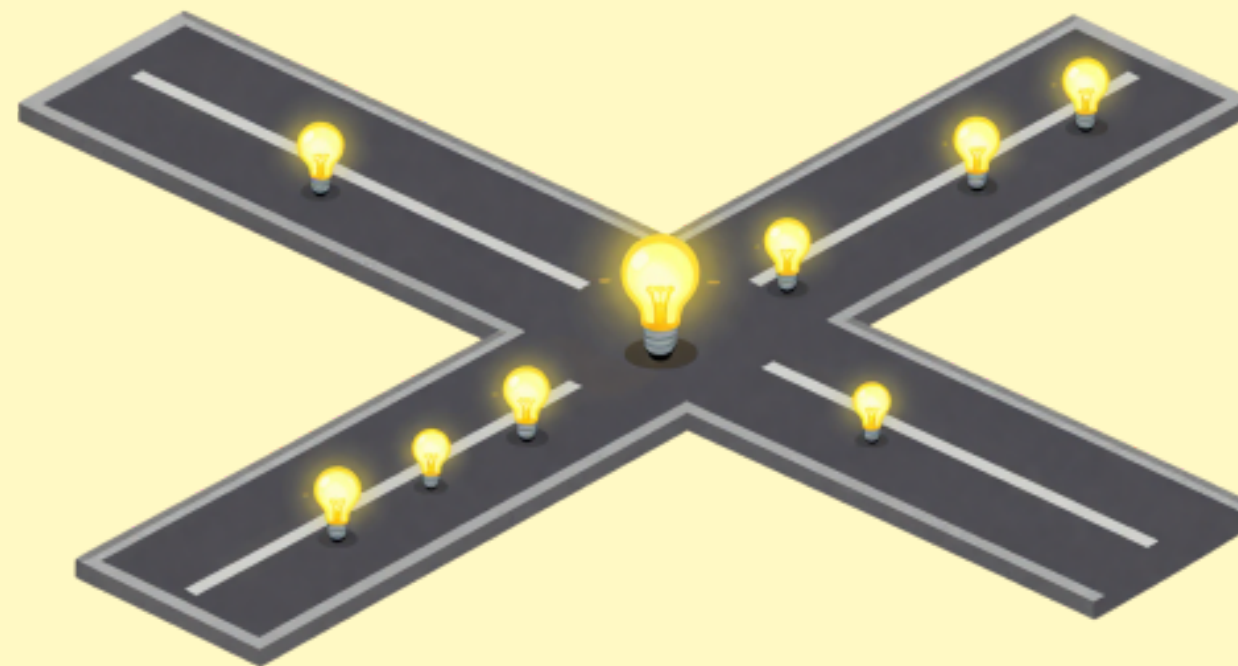


How Does It Look?

Imagine a road splitting into lanes.

Each lane has its own LED.

All lanes start and end at the
same point (the battery).

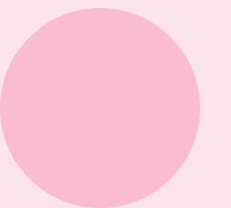
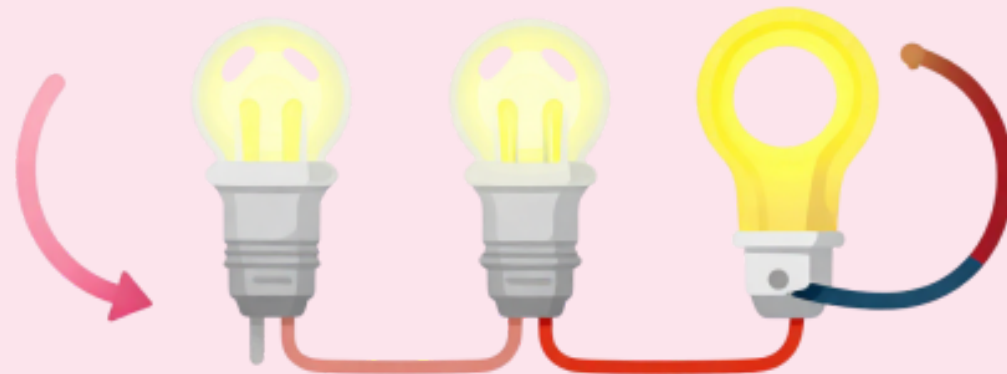


Independent LEDs

In a parallel circuit, each LED works on its own.

If one LED burns out, the others keep glowing!

Each LED gets the FULL battery voltage.

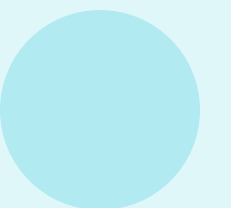
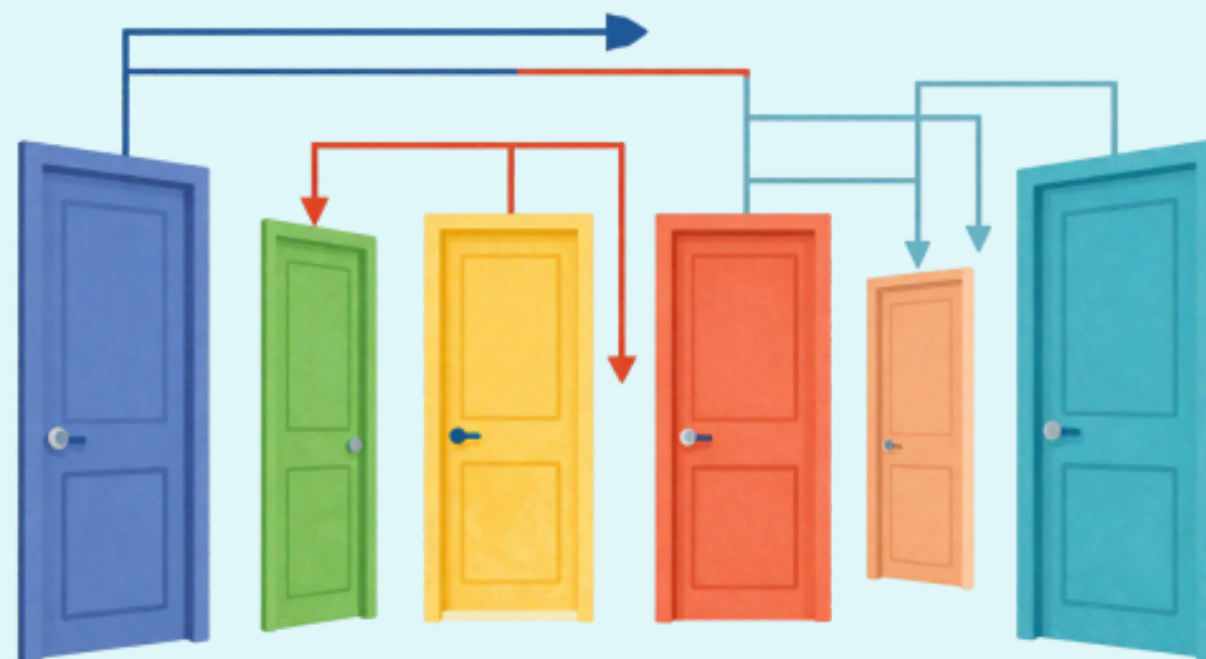


Why Independent?

Each LED has its own complete path.

Breaking one path does NOT break the others.

Think of it like doors in a hallway — closing one doesn't block the rest!

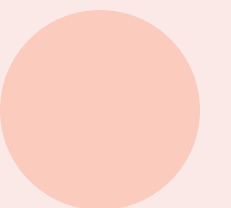


Real Life Example

Your home lights are in parallel!

When you switch off the kitchen light,
the bedroom light stays on.

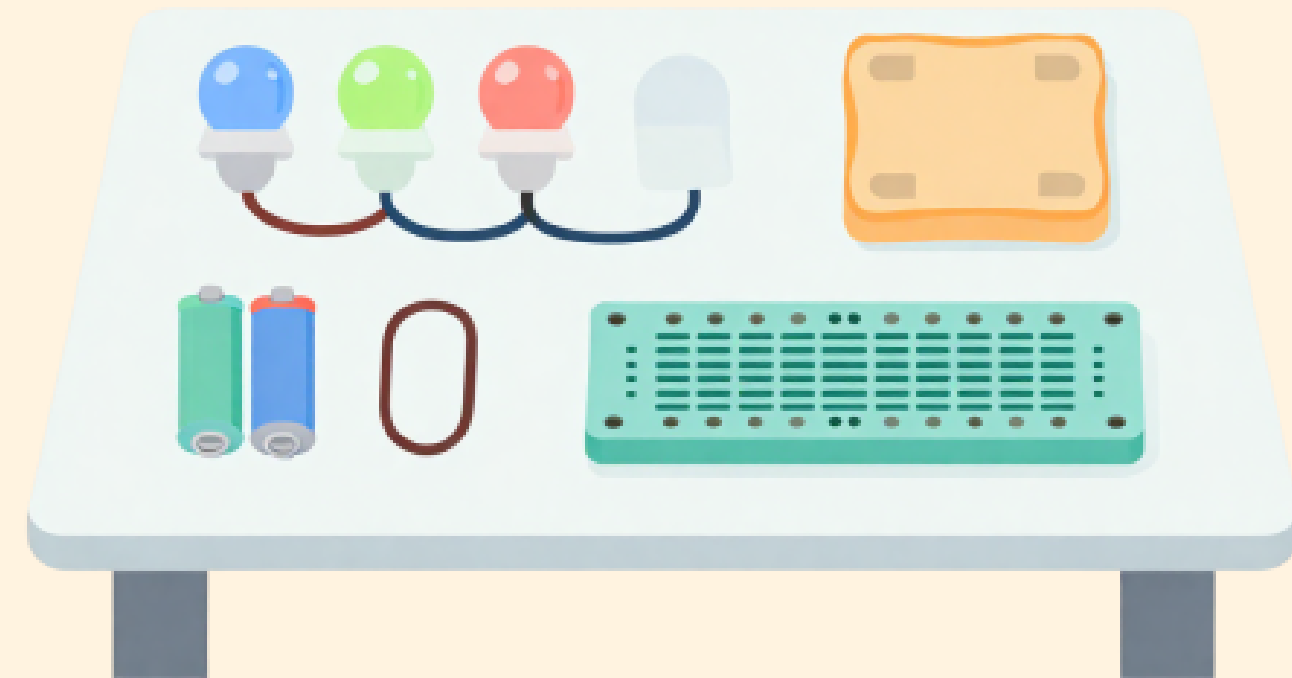
Each room has its own circuit path.



Let's Build One!

Materials needed:

- 1 battery (3V or 4.5V)
- 3 LEDs (different colours)
- Wires
- A breadboard (optional)



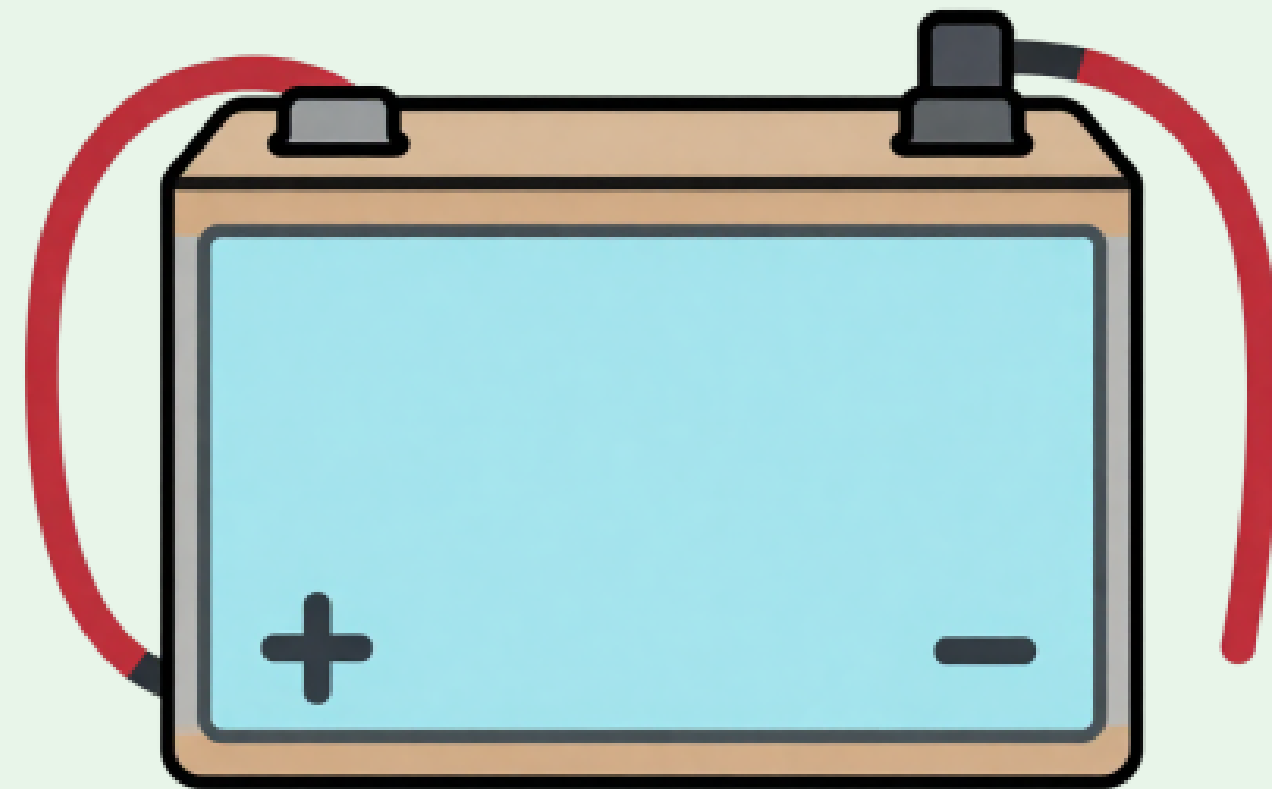
Step 1: Connect the Battery

Place the battery in the holder.

Identify the positive (+) and negative (-) terminals.

Red wire = positive

Black wire = negative

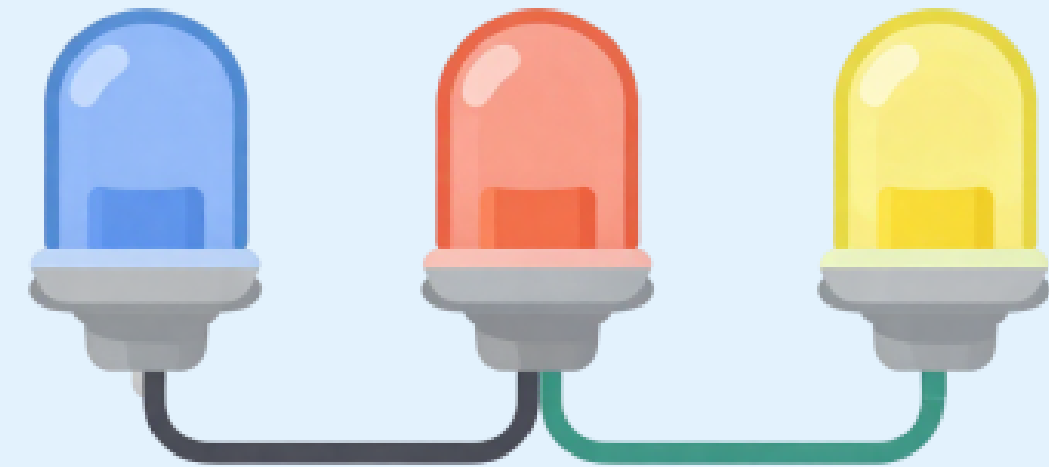


Step 2: Wire Each LED

Connect each LED's long leg (+)
to the positive wire.

Connect each LED's short leg (-)
to the negative wire.

All LEDs share the same two main wires!

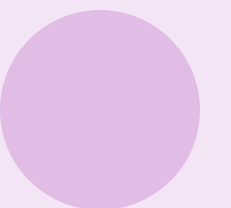
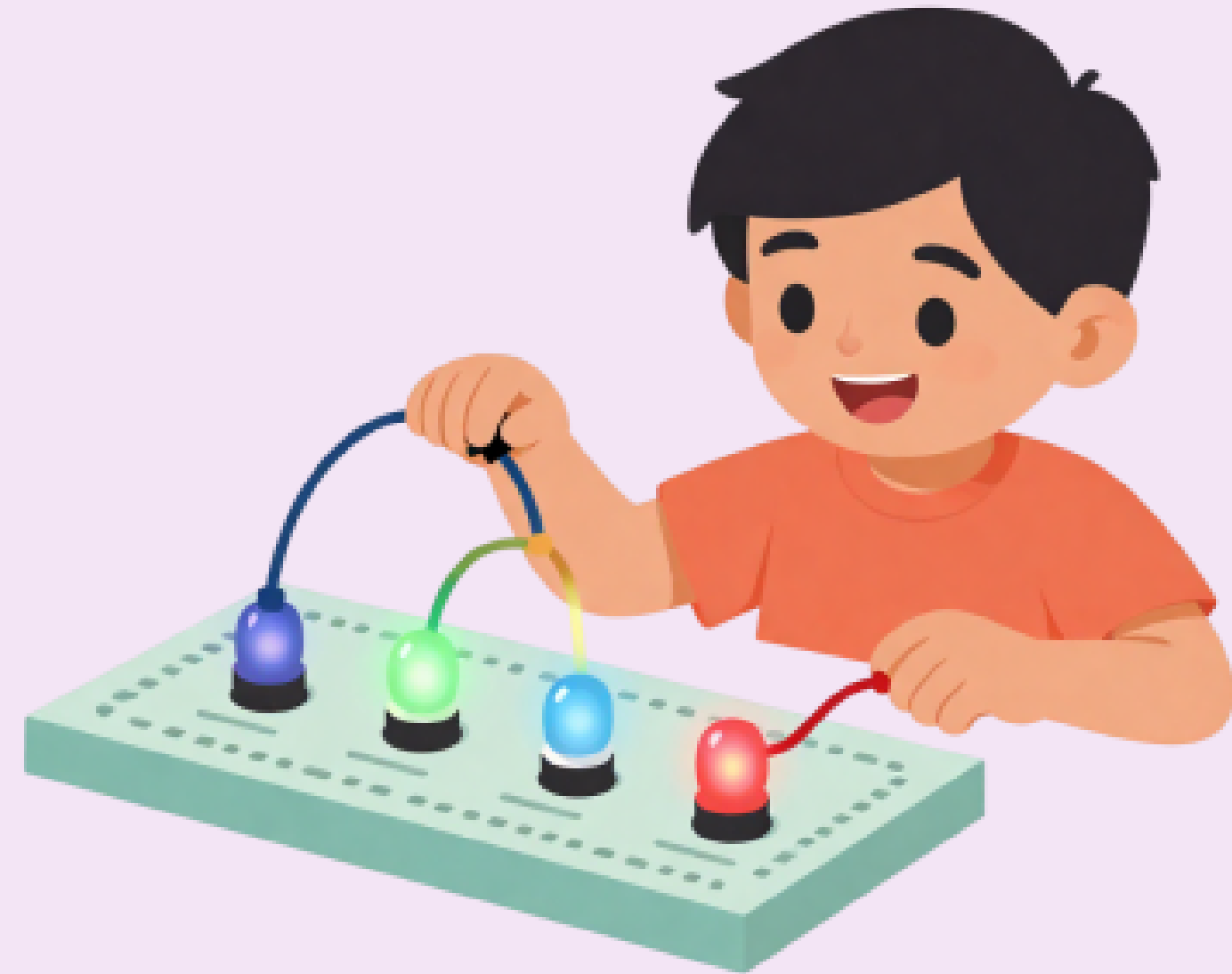


Step 3: Test It!

Turn on the circuit.
All 3 LEDs should glow!

Now remove one LED.
Do the other two still glow? YES!

That's a parallel circuit working!



Series vs Parallel

Series:

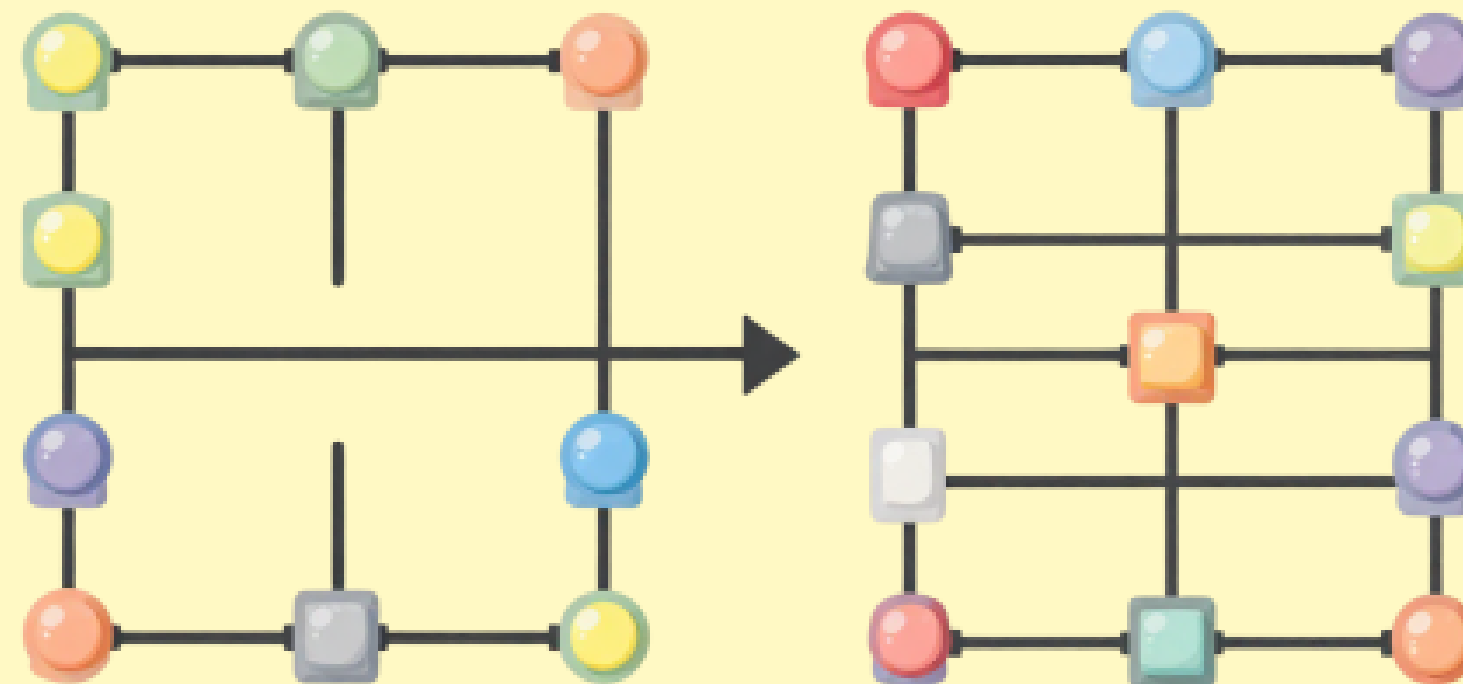
One path | LEDs share voltage

One breaks = all off

Parallel:

Many paths | Full voltage each

One breaks = others stay on

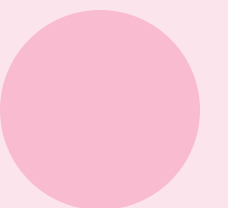


Fun Fact!

Traffic lights use parallel circuits!

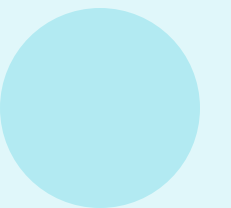
Red, yellow, and green LEDs work independently.

If one colour fails, the others still work to keep us safe.



Quick Quiz!

1. How many paths does a parallel circuit have?
2. What happens if one LED is removed?
3. Do home lights use series or parallel?
4. Which leg of LED is positive – long or short?



Great Job!

You learned about
parallel circuits!

Now you know why your
home lights work independently.

